

CompTIA.

A+ Core 2

Exam 220-1202

A+ 220-1202

Module 13



Managing Windows

Learning Outcomes

Prepare for A+ Core 2 by:

- Using management consoles
- Using performance and troubleshooting tools
- Using command-line tools

Lesson 13.1

Use Management Consoles

Device Manager (Slide 1 of 3)

1

Updating devices

- Driver information

2

Troubleshooting malfunctioning devices

- Built-in troubleshooter to assist in diagnosis and correction of issues

3

Removing devices

4

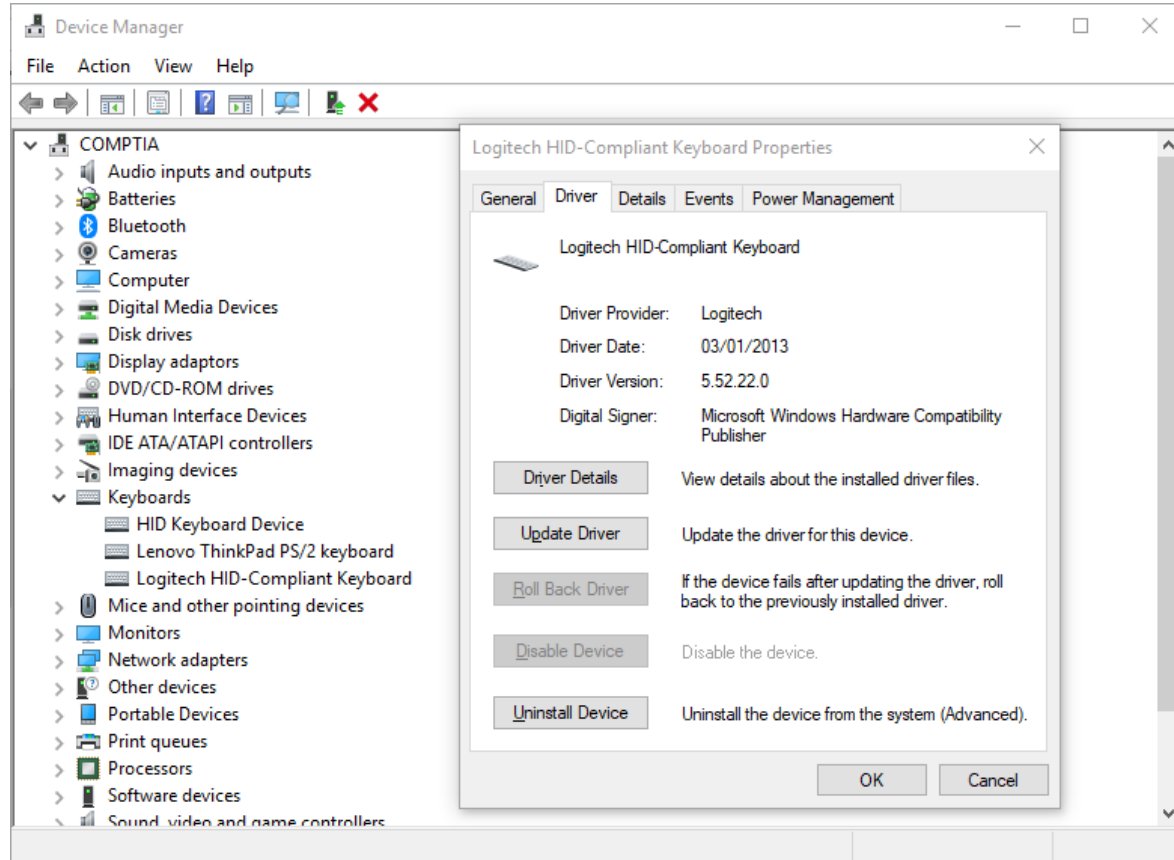
Uninstalling devices

5

Disabling devices

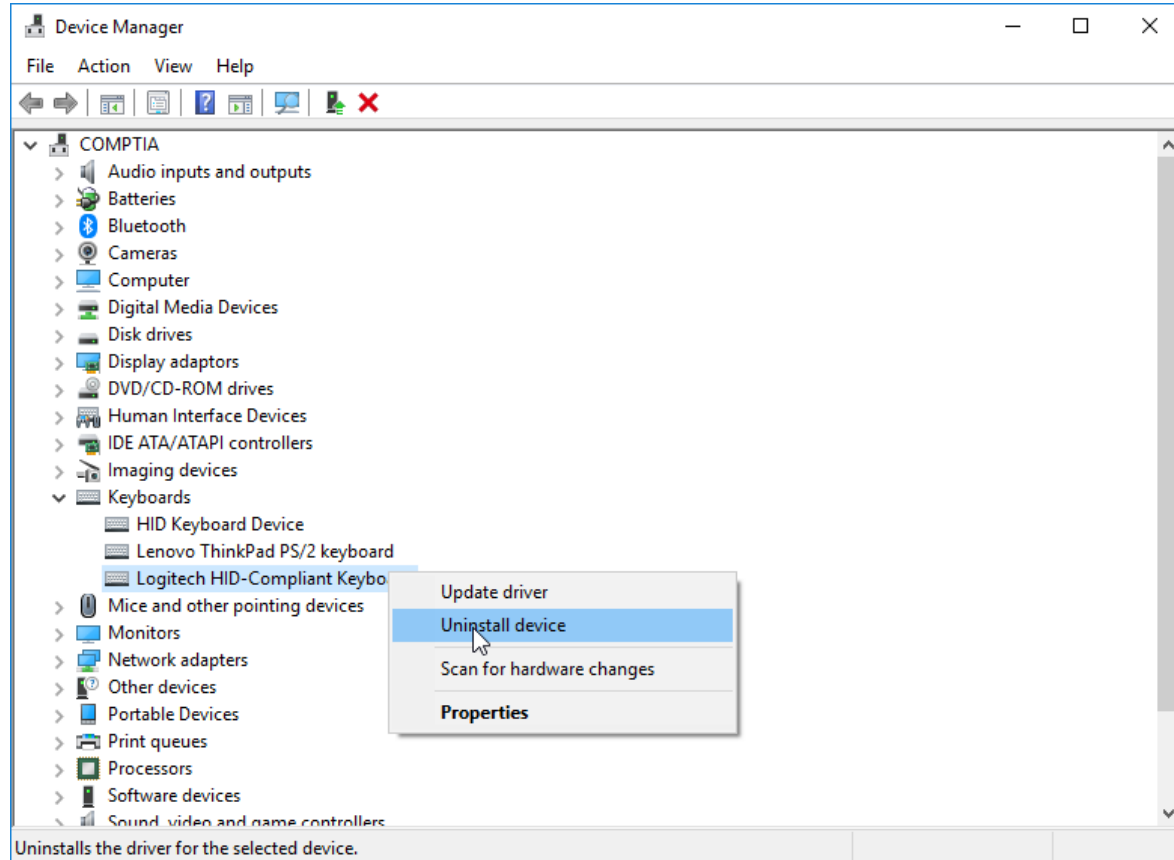
- Some hardware devices can be disabled in the BIOS/UEFI menus

Device Manager (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Device Manager (Slide 3 of 3)



Screenshot courtesy of Microsoft.

Disk Management Console (Slide 1 of 2)



Manage fixed and removable disks and partitions



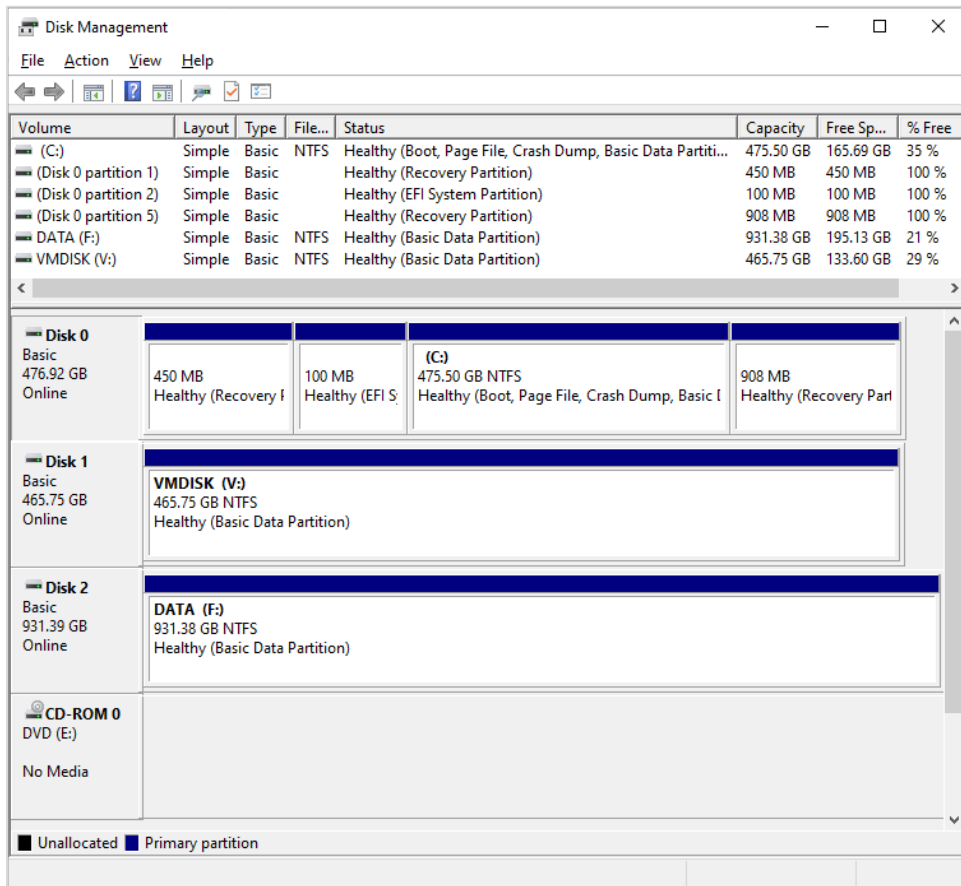
HDD, SSD, and optical drives



Disk management tasks

Initialize
Partition
Format
Repartition

Disk Management Console (Slide 2 of 2)



The screenshot displays the Windows Disk Management console. At the top, a table lists the volumes on the system. Below this, the details for each disk are shown, including the disk type, capacity, and the status of its partitions. A legend at the bottom indicates that black represents unallocated space and blue represents primary partitions.

Volume	Layout	Type	File...	Status	Capacity	Free Sp...	% Free
(C:)	Simple	Basic	NTFS	Healthy (Boot, Page File, Crash Dump, Basic Data Partiti...	475.50 GB	165.69 GB	35 %
(Disk 0 partition 1)	Simple	Basic		Healthy (Recovery Partition)	450 MB	450 MB	100 %
(Disk 0 partition 2)	Simple	Basic		Healthy (EFI System Partition)	100 MB	100 MB	100 %
(Disk 0 partition 3)	Simple	Basic		Healthy (Recovery Partition)	908 MB	908 MB	100 %
DATA (F:)	Simple	Basic	NTFS	Healthy (Basic Data Partition)	931.38 GB	195.13 GB	21 %
VMDISK (V:)	Simple	Basic	NTFS	Healthy (Basic Data Partition)	465.75 GB	133.60 GB	29 %

Disk	Capacity	Status	Partitions	
Disk 0 Basic 476.92 GB Online	450 MB Healthy (Recovery I	100 MB Healthy (EFI S	(C:) 475.50 GB NTFS Healthy (Boot, Page File, Crash Dump, Basic I	908 MB Healthy (Recovery Part
Disk 1 Basic 465.75 GB Online	VMDISK (V:) 465.75 GB NTFS Healthy (Basic Data Partition)			
Disk 2 Basic 931.39 GB Online	DATA (F:) 931.38 GB NTFS Healthy (Basic Data Partition)			
CD-ROM 0 DVD (E:)	No Media			

■ Unallocated ■ Primary partition

Screenshot courtesy of Microsoft.



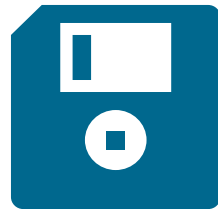
Disk DeFragmenter

HDD

- Reorganizes data into contiguous clusters to improve access timelines

SSD

- Performs a trim operation to reduce cell degradation

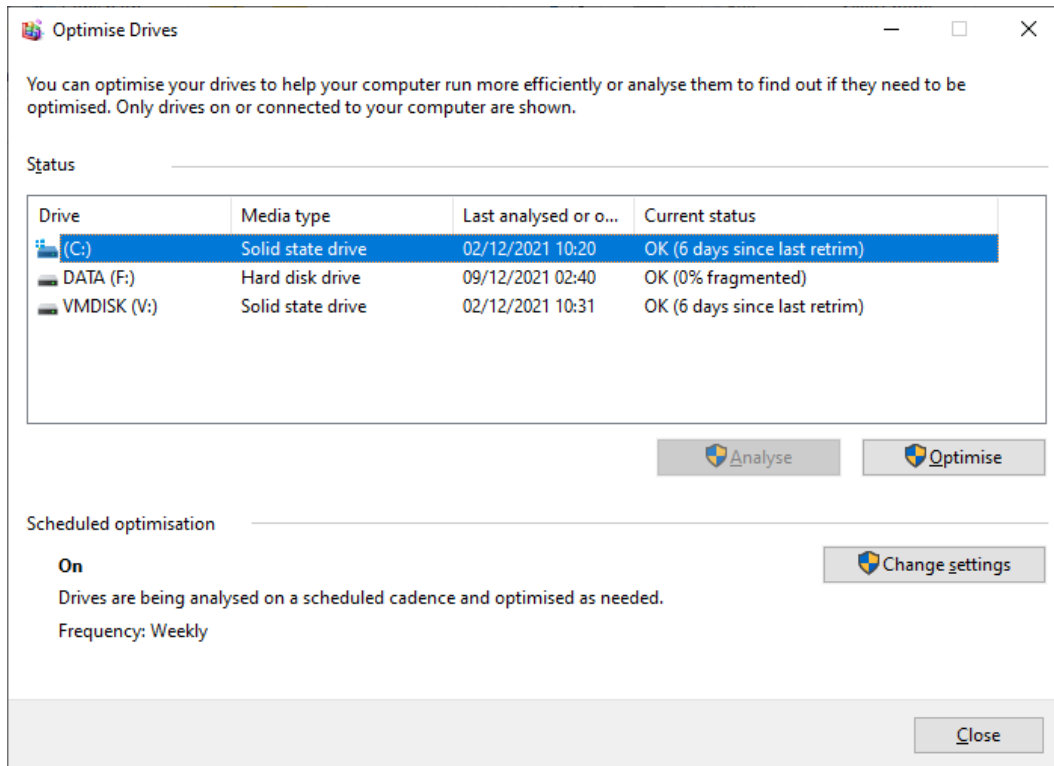


Disk Clean-up

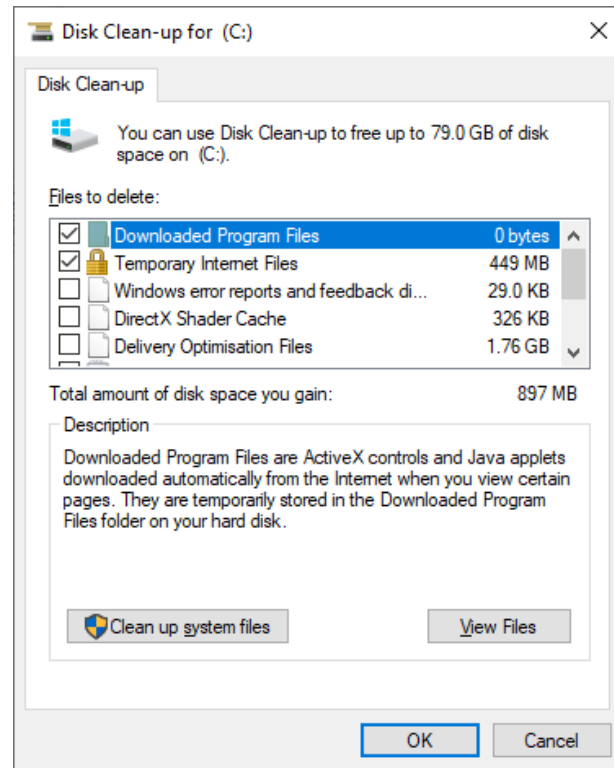
Removes temporary files and files marked for deletion

Can remove old Windows installation files and updates no longer needed by the OS

Disk Maintenance Tools (slide 2 of 2)



Screenshot courtesy of Microsoft.

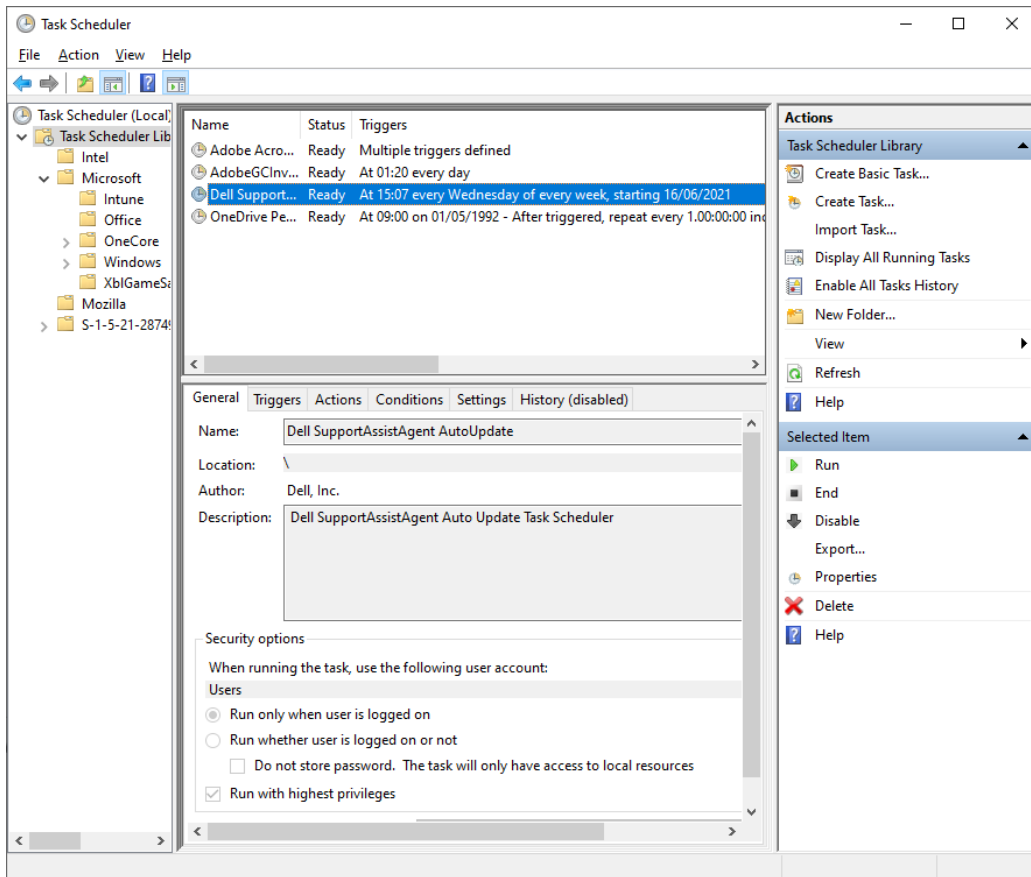


Screenshot courtesy of Microsoft.

Task Scheduler (Slide 1 of 2)

- Automates operations run at specific intervals or based on a trigger
- Logging
 - Allows troubleshooting when task failure
- Schedules tasks
 - Updates, backups, and synchronizing data

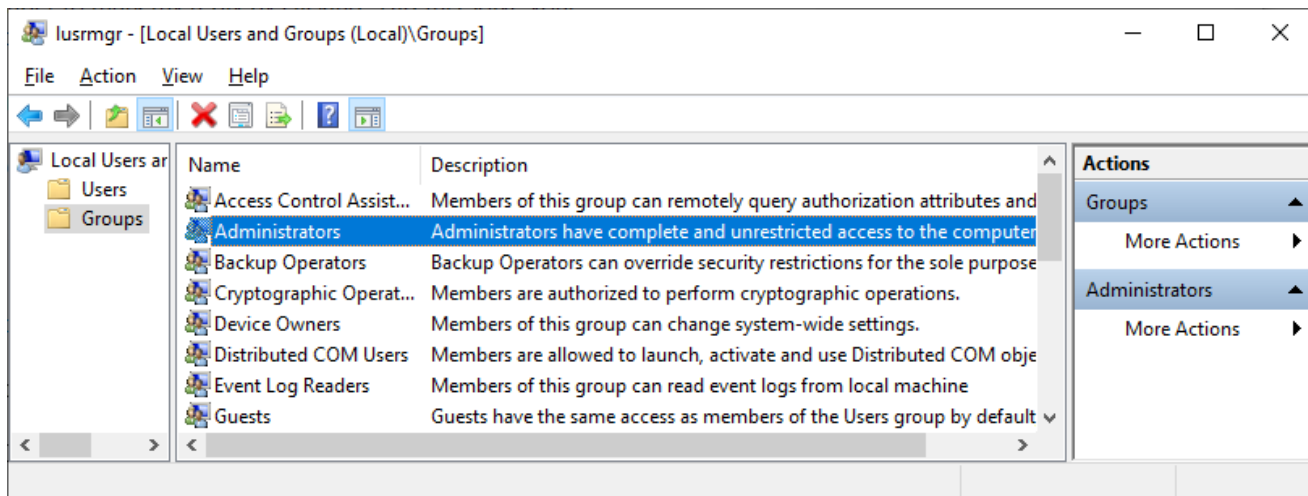
Task Scheduler (Slide 2 of 2)



Screenshot courtesy of Microsoft.

Local Users and Groups Console

- Advanced interface for managing user accounts on the local system
- Create and modify security groups for easier permissions management



Certificate Manager (Slide 1 of 2)



Manage digital certificates for identify of users, computers, or services

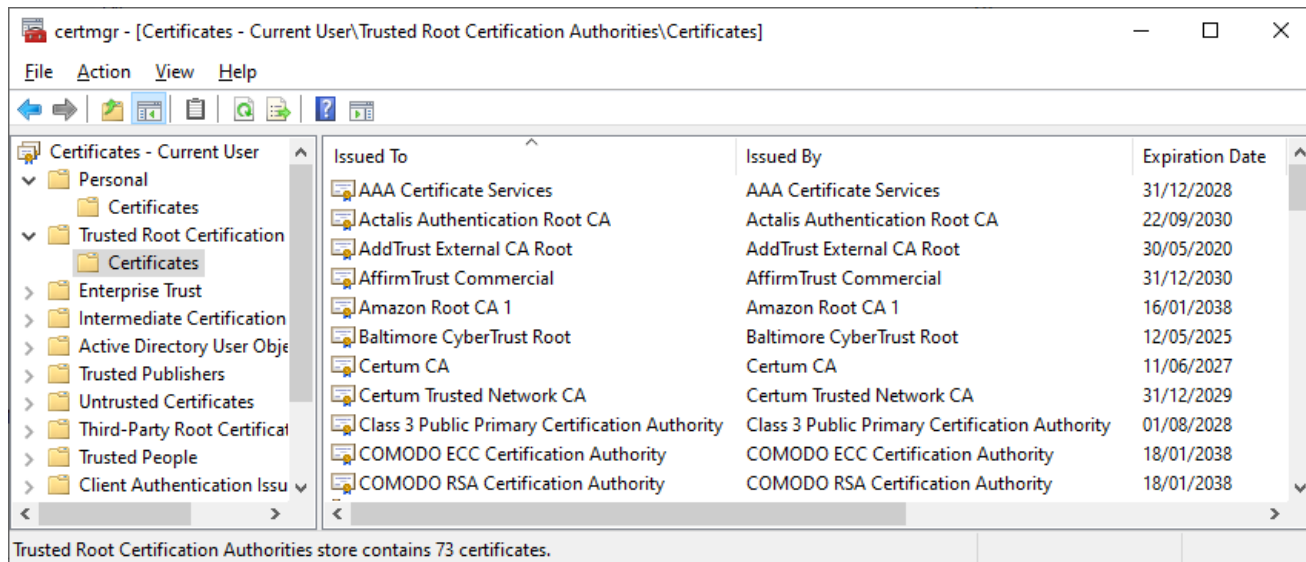


Facilitates verification of certificate information



Certmgr.msc (current user) or certlm.msc (computer certificates)

Certificate Manager (Slide 2 of 2)



Screenshot courtesy of Microsoft.

01

Manage local policies for users and computers

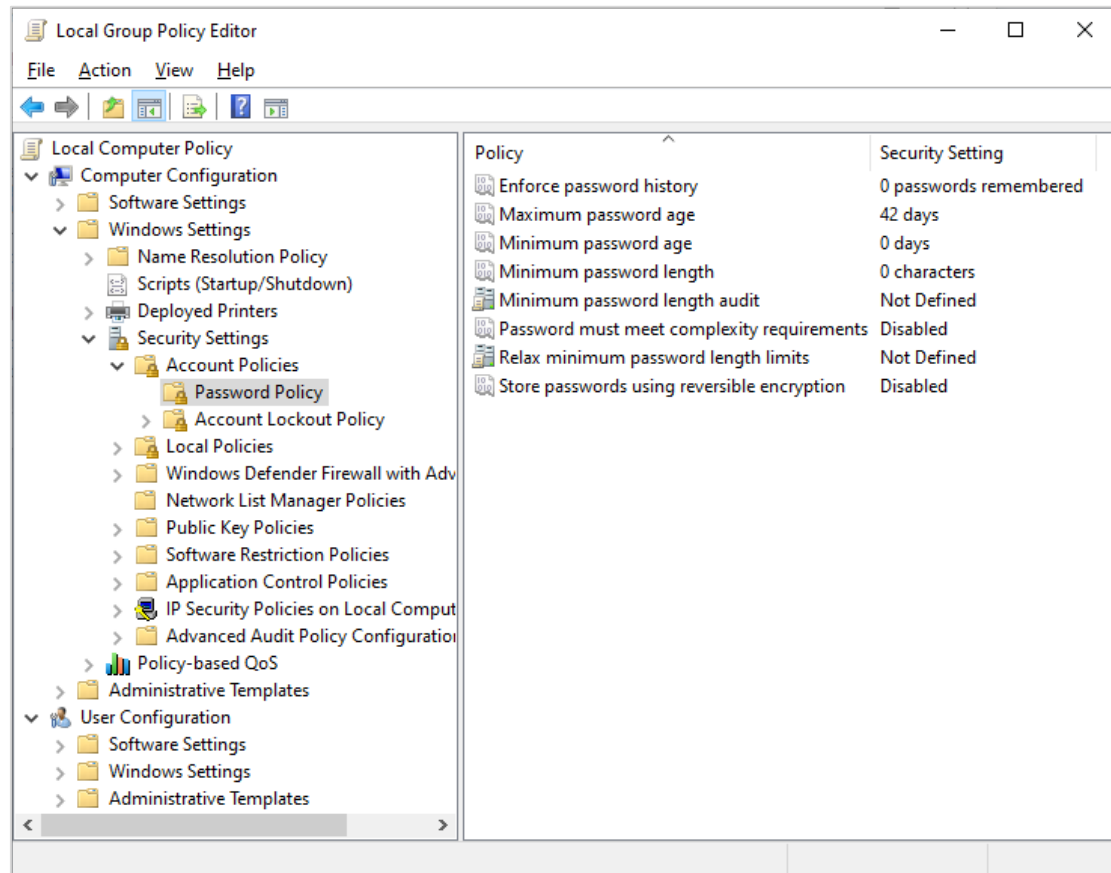
02

Configure Windows settings without direct registry changes

03

Apply settings efficiently system-wide

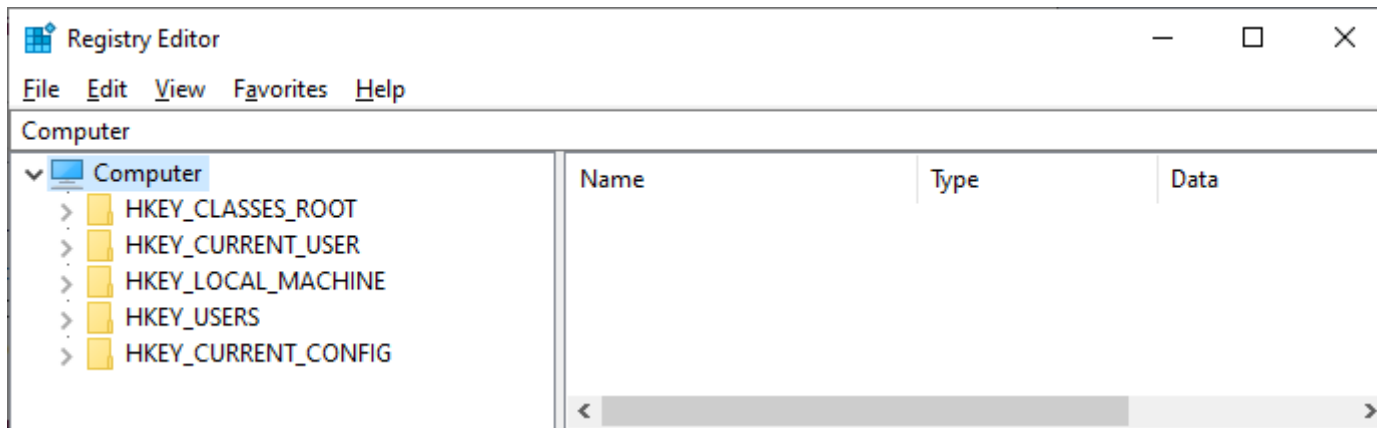
Group Policy Editor (Slide 2 of 2)



Screenshot courtesy of Microsoft.

Registry Editor (Slide 1 of 2)

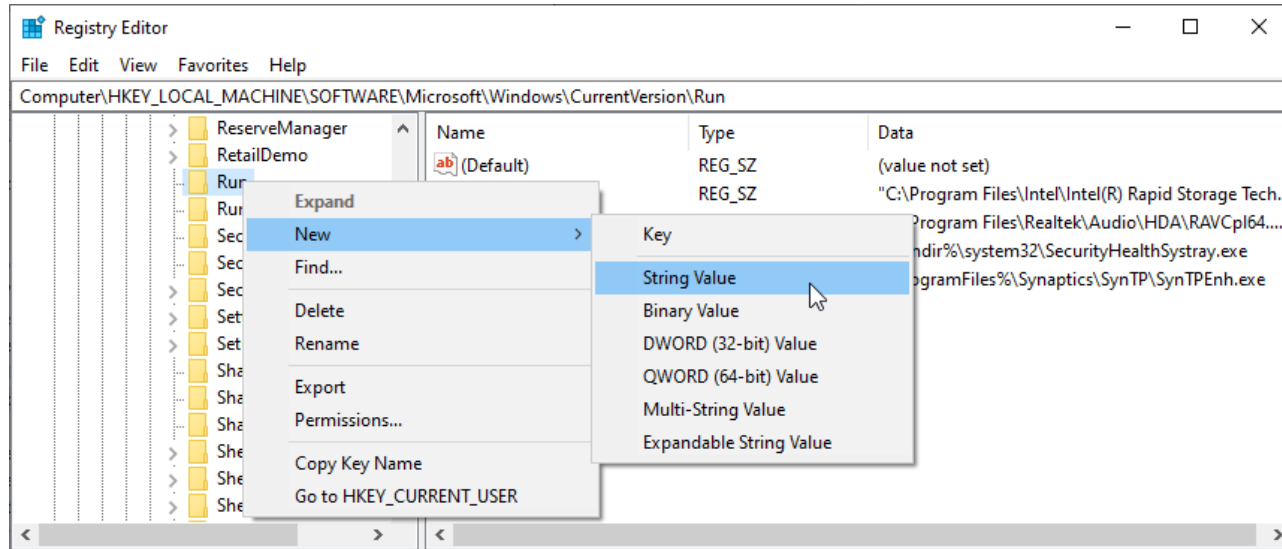
- Five root keys



Screenshot courtesy of Microsoft.

Registry Editor (Slide 2 of 2)

- Edit settings and configurations of user accounts, hardware, applications, and operating system



Screenshot courtesy of Microsoft.

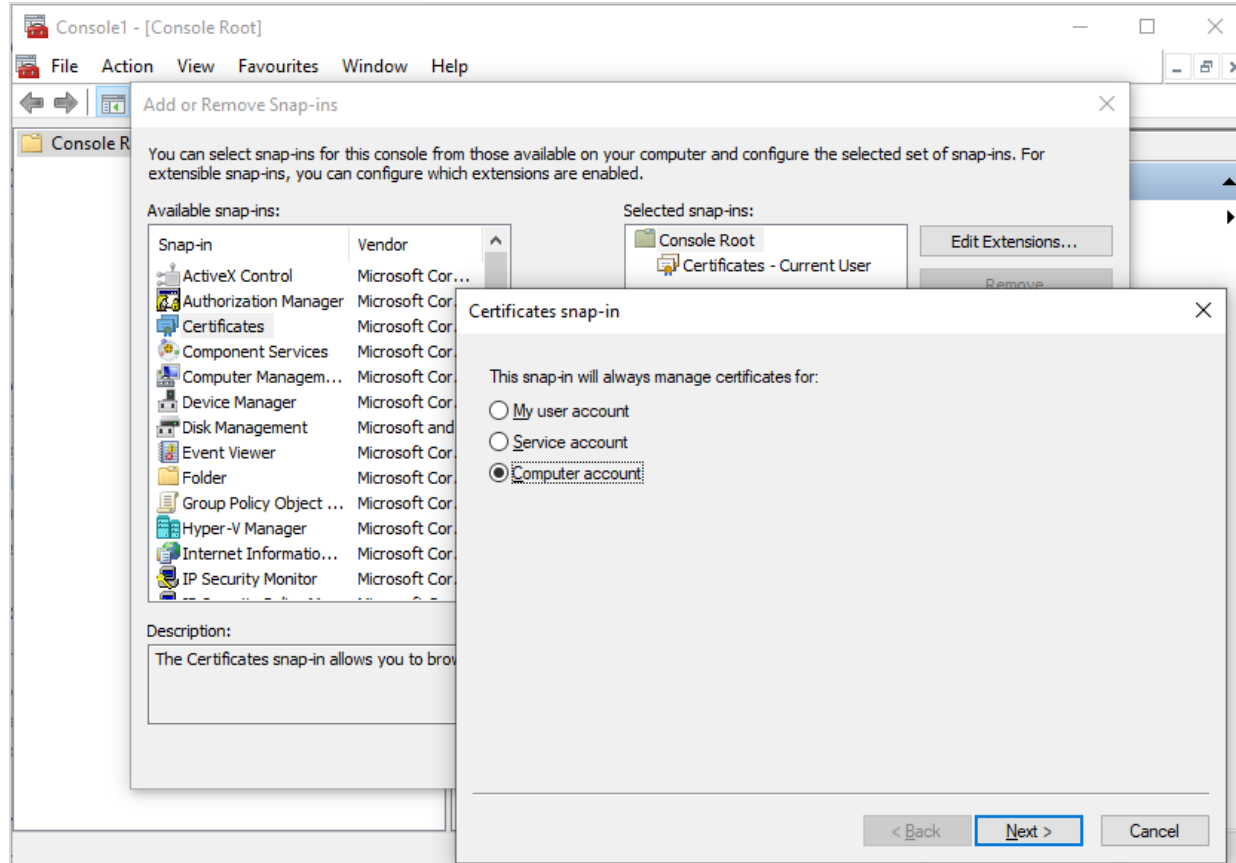


Container for various snap-ins



Place all necessary snap-ins
needed for job in one location

Custom Microsoft Management Consoles (Slide 2 of 2)



Screenshot courtesy of Microsoft.

Lab Activity

- Lab: Disk Management
(available in CertMaster Learn and CertMaster Perform)
- Lab: Users and Groups
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Manage User Accounts
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Management Consoles

- Device Manager
- Disk Management Console
- Disk Maintenance Tools
- Task Scheduler
- Local Users and Groups Console
- Certificate Manager
- Group Policy Editor
- Registry Editor
- Custom Microsoft Management Consoles

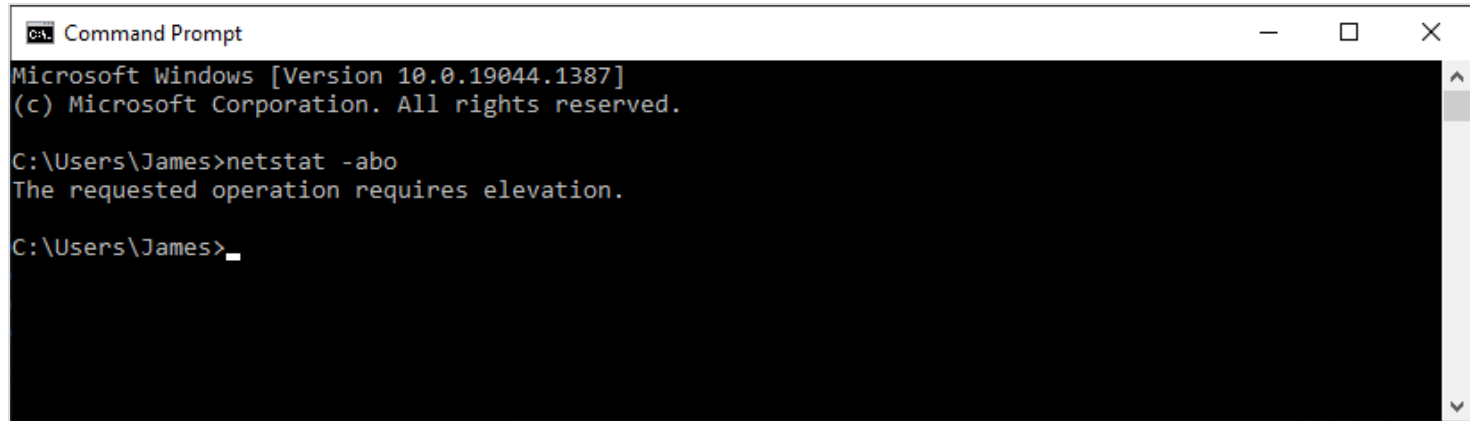
Lesson 13.2

Command-Line Tools

Command Prompt (Slide 1 of 2)

- "Run As Administrator" option for elevated permissions
- Syntax
 - order of command, switches, and arguments
- CTRL+C
 - stop command execution
- Help
 - *command /?* Or *help command*

Command Prompt (Slide 2 of 2)



```
CA Command Prompt
Microsoft Windows [Version 10.0.19044.1387]
(c) Microsoft Corporation. All rights reserved.

C:\Users\James>netstat -abo
The requested operation requires elevation.

C:\Users\James>_
```

Screenshot courtesy of Microsoft.

Navigation Commands (Slide 1 of 2)



Dir

/o: (n, s, e, d)

/t: (c, a, w)

/a: (r, h, s, a)



Cd

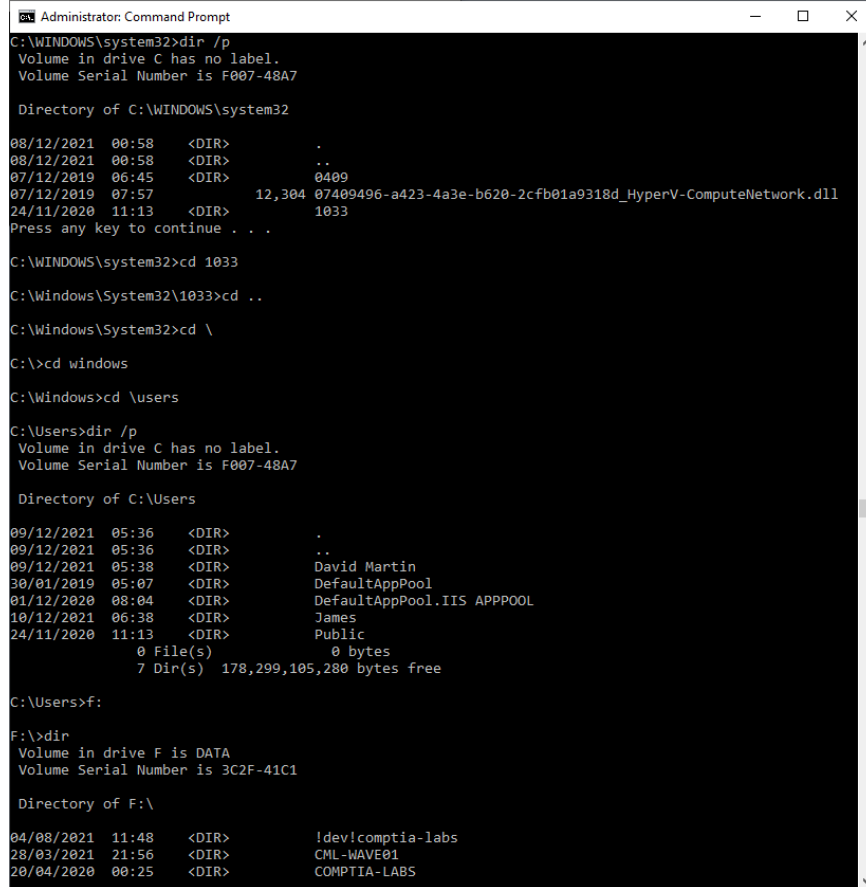
..
\\



Change drive

Type drive letter then : (D: or E:)

Navigation Commands (Slide 2 of 2)



```
Administrator: Command Prompt
C:\WINDOWS\system32>dir /p
Volume in drive C has no label.
Volume Serial Number is F007-48A7

Directory of C:\WINDOWS\system32

08/12/2021  00:58  <DIR>      .
08/12/2021  00:58  <DIR>      ..
07/12/2019  06:45  <DIR>      0409
07/12/2019  07:57      12,304  07409496-a423-4a3e-b620-2cfb01a9318d_HyperV-ComputeNetwork.dll
24/11/2020  11:13  <DIR>      1033
Press any key to continue . . .

C:\WINDOWS\system32>cd 1033

C:\Windows\System32\1033>cd ..

C:\Windows\System32>cd \

C:\>cd windows

C:\Windows>cd \users

C:\Users>dir /p
Volume in drive C has no label.
Volume Serial Number is F007-48A7

Directory of C:\Users

09/12/2021  05:36  <DIR>      .
09/12/2021  05:36  <DIR>      ..
09/12/2021  05:38  <DIR>      David Martin
30/01/2019  05:07  <DIR>      DefaultAppPool
01/12/2020  08:04  <DIR>      DefaultAppPool.IIS APPPOOL
10/12/2021  06:38  <DIR>      James
24/11/2020  11:13  <DIR>      Public
                0 File(s)      0 bytes
                7 Dir(s)  178,299,105,280 bytes free

C:\Users>F:

F:\>dir
Volume in drive F is DATA
Volume Serial Number is 3C2F-41C1

Directory of F:\

04/08/2021  11:48  <DIR>      !dev!comptia-labs
28/03/2021  21:56  <DIR>      CML-WAVE01
20/04/2020  00:25  <DIR>      COMPTIA-LABS
```

Screenshot courtesy of Microsoft.

File Management Commands

- move
- copy
- robocopy
 - Recommended for long filenames and NTFS attribute handling
- md or mkdir
- rd or rmdir

Disk Management Commands (Slide 1 of 2)

- Diskpart
 - command line Disk Management tool
 - select disk x
 - detail disk
 - select partition x
 - assign, delete, extend
 - exit
- format
 - Erases and reformats the drive
- chkdsk
 - Scans file system and sectors for faults and attempts to repair

Disk Management Commands (slide 2 of 2)

```
Administrator: Command Prompt - diskpart
DISKPART> select disk 0

Disk 0 is now the selected disk.

DISKPART> detail disk

NVMe SAMSUNG MZVPV512
Disk ID: {59562218-3300-452F-B899-2B6CF427BD10}
Type : NVMe
Status : Online
Path : 0
Target : 0
LUN ID : 0
Location Path : PCIR00T(0)#PCI(1B00)#PCI(0000)#NVME(P00T00L00)
Current Read-only State : No
Read-only : No
Boot Disk : Yes
Pagefile Disk : Yes
Hibernation File Disk : No
Crashdump Disk : Yes
Clustered Disk : No

  Volume ###  Ltr  Label       Fs      Type        Size      Status       Info
  -----
  Volume 1      C           NTFS      Partition   475 GB     Healthy      Boot
  Volume 2             Recovery  NTFS      Partition   450 MB     Healthy      Hidden
  Volume 3             FAT32     Partition   100 MB     Healthy      System
  Volume 4             NTFS      Partition   908 MB     Healthy      Hidden

DISKPART> select volume 1

Volume 1 is the selected volume.

DISKPART> detail volume

  Disk ###  Status       Size      Free      Dyn  Gpt
  -----
  * Disk 0   Online             476 GB    1024 KB          *

Read-only           : No
Hidden              : No
No Default Drive Letter: No
Shadow Copy         : No
Offline             : No
BitLocker Encrypted : No
Installable         : Yes

Volume Capacity    : 475 GB
Volume Free Space   : 166 GB

DISKPART>
```

Screenshot courtesy of Microsoft.

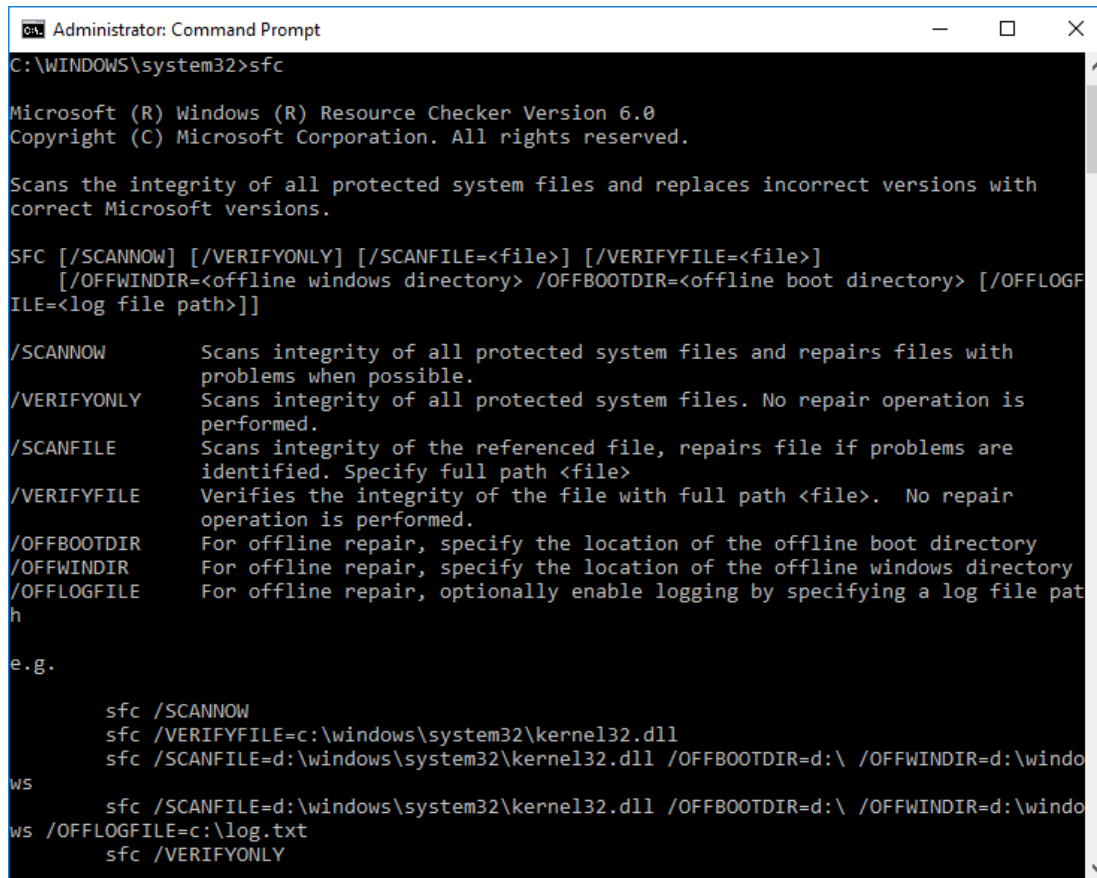
System Management Commands (Slide 1 of 3)

- shutdown
 - /s shutdown
 - /h hibernate
 - /l log off
 - /r restart
 - /t assign a delay to the shutdown command execution
 - Example: shutdown /r /t 45 will restart the system after 45 seconds

System Management Commands (Slide 2 of 3)

- `sfc`
 - System file checker to fix corrupt or damaged system files
- `winver`
 - Displays Windows version information
- `whoami`
 - Displays current user who is logged in

System Management Commands (Slide 3 of 3)



```
Administrator: Command Prompt
C:\WINDOWS\system32>sfc

Microsoft (R) Windows (R) Resource Checker Version 6.0
Copyright (C) Microsoft Corporation. All rights reserved.

Scans the integrity of all protected system files and replaces incorrect versions with
correct Microsoft versions.

SFC [/SCANNOW] [/VERIFYONLY] [/SCANFILE=<file>] [/VERIFYFILE=<file>]
    [/OFFWINDIR=<offline windows directory> /OFFBOOTDIR=<offline boot directory> [/OFFLOGFI
    LE=<log file path>]]

/SCANNOW          Scans integrity of all protected system files and repairs files with
                  problems when possible.
/VERIFYONLY       Scans integrity of all protected system files. No repair operation is
                  performed.
/SCANFILE         Scans integrity of the referenced file, repairs file if problems are
                  identified. Specify full path <file>
/VERIFYFILE       Verifies the integrity of the file with full path <file>. No repair
                  operation is performed.
/OFFBOOTDIR       For offline repair, specify the location of the offline boot directory
/OFFWINDIR        For offline repair, specify the location of the offline windows directory
/OFFLOGFILE       For offline repair, optionally enable logging by specifying a log file pat
h

e.g.

sfc /SCANNOW
sfc /VERIFYFILE=c:\windows\system32\kernel32.dll
sfc /SCANFILE=d:\windows\system32\kernel32.dll /OFFBOOTDIR=d:\ /OFFWINDIR=d:\windo
ws
sfc /SCANFILE=d:\windows\system32\kernel32.dll /OFFBOOTDIR=d:\ /OFFWINDIR=d:\windo
ws /OFFLOGFILE=c:\log.txt
sfc /VERIFYONLY
```

Screenshot courtesy of Microsoft.

Lab Activity

- Lab: Manage Files and Folders
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Configure File Storage
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Command-Line Tools

- Command Prompt
- Navigation Commands
- File Management Commands
- Disk Management Commands
- System Management Commands

Lesson 13.3

Windows Networking

Windows Network Connection Types (Slide 1 of 3)

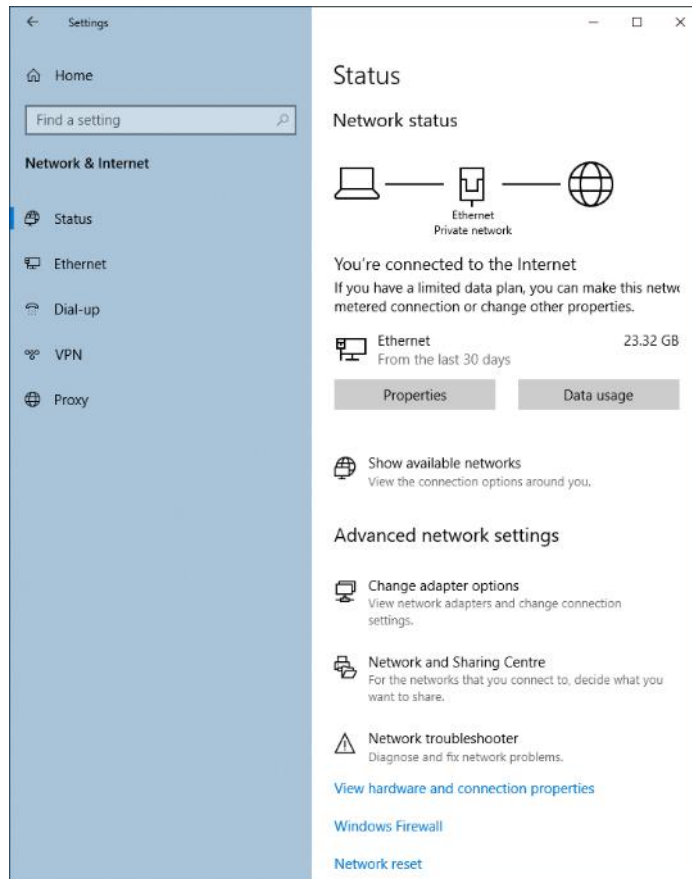
Wired

- EthernetX, where x is the number of the interface
- Configure by navigating to Device Manager > Properties > Advanced tab

Wireless

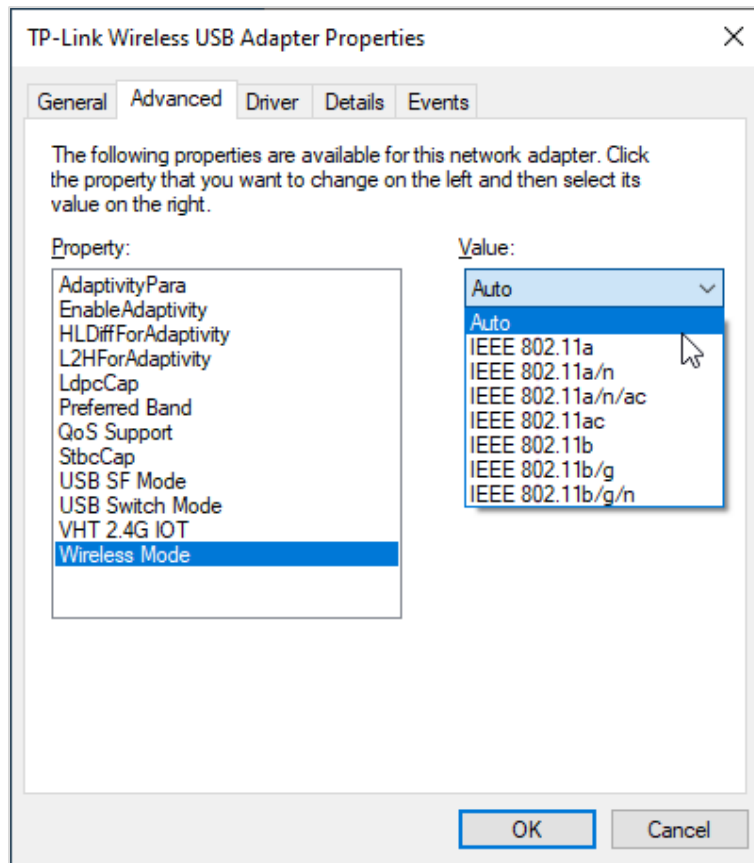
- Scan for network or enter SSID manually
- Connect automatically selection
 - Only for trusted networks

Windows Network Connection Types (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Windows Network Connection Types (Slide 3 of 3)



Screenshot courtesy of Microsoft.

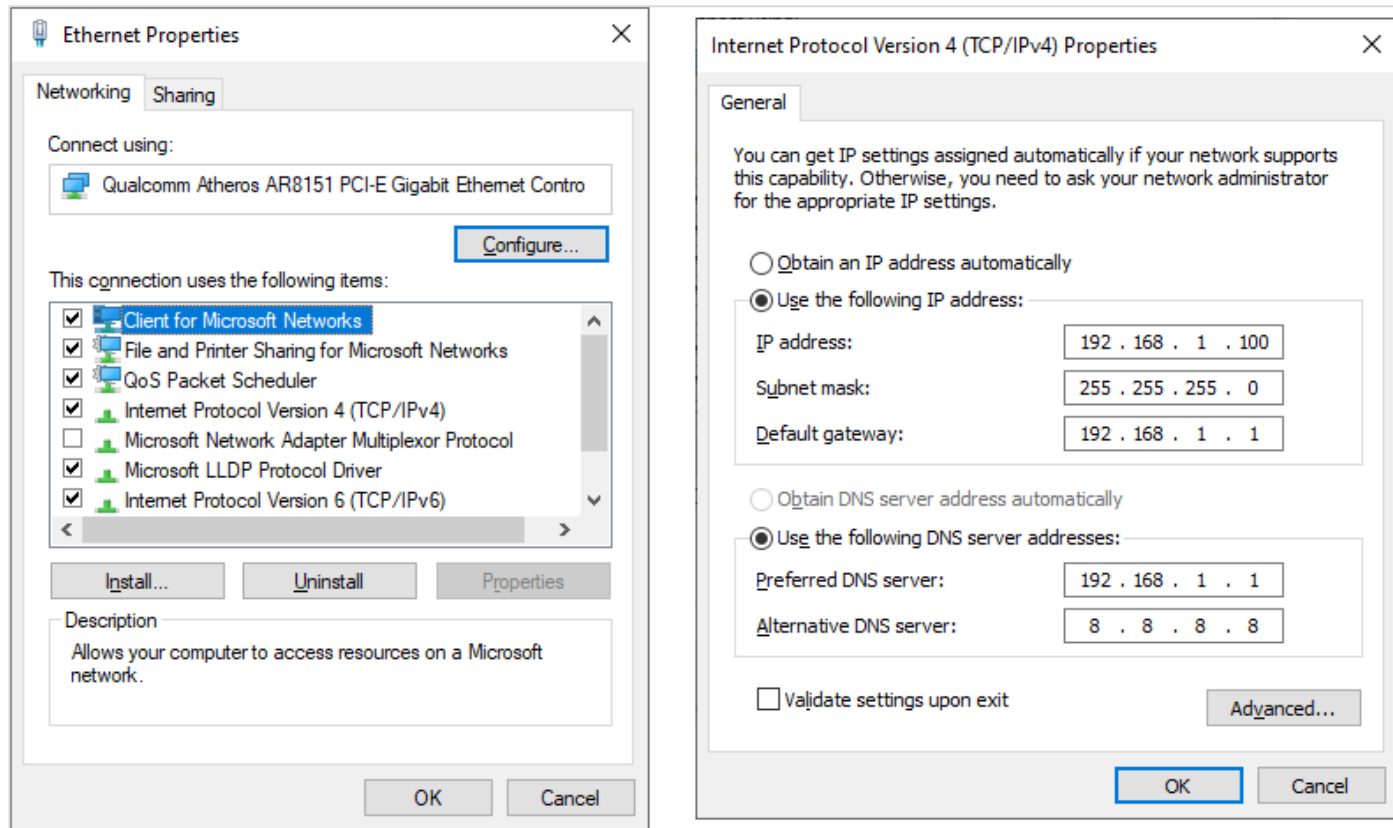
IP Addressing Schemes

- Configuration requirements
 - IP address
 - v4- 32-bit address
 - dotted decimal
 - v6- 128-bit address
 - hex notation
 - Subnet mask
 - Default gateway
 - DNS server IP
- Static vs dynamic (DHCP)

Windows Client Configuration (Slide 1 of 3)

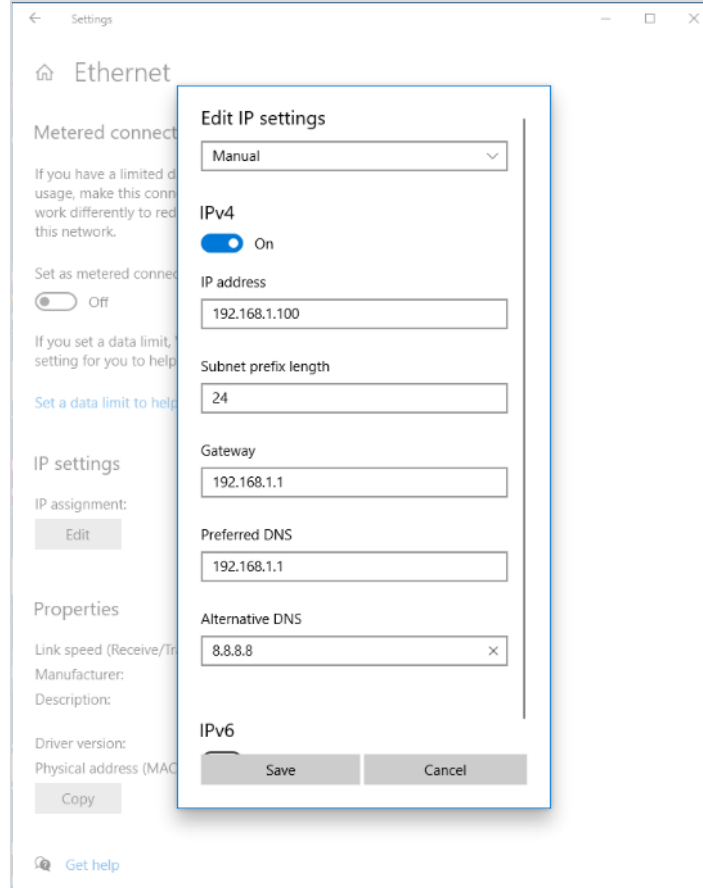
- Network & Internet settings or Network Connections applet
- DHCP is default configuration for adapters
- Installed by default
 - Microsoft Networks
 - File and Printer Sharing for Microsoft Networks
 - IPv4 and IPv6
 - Link-layer Topology Discovery
 - Network mapping and discovery function

Windows Client Configuration (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Windows Client Configuration (Slide 3 of 3)

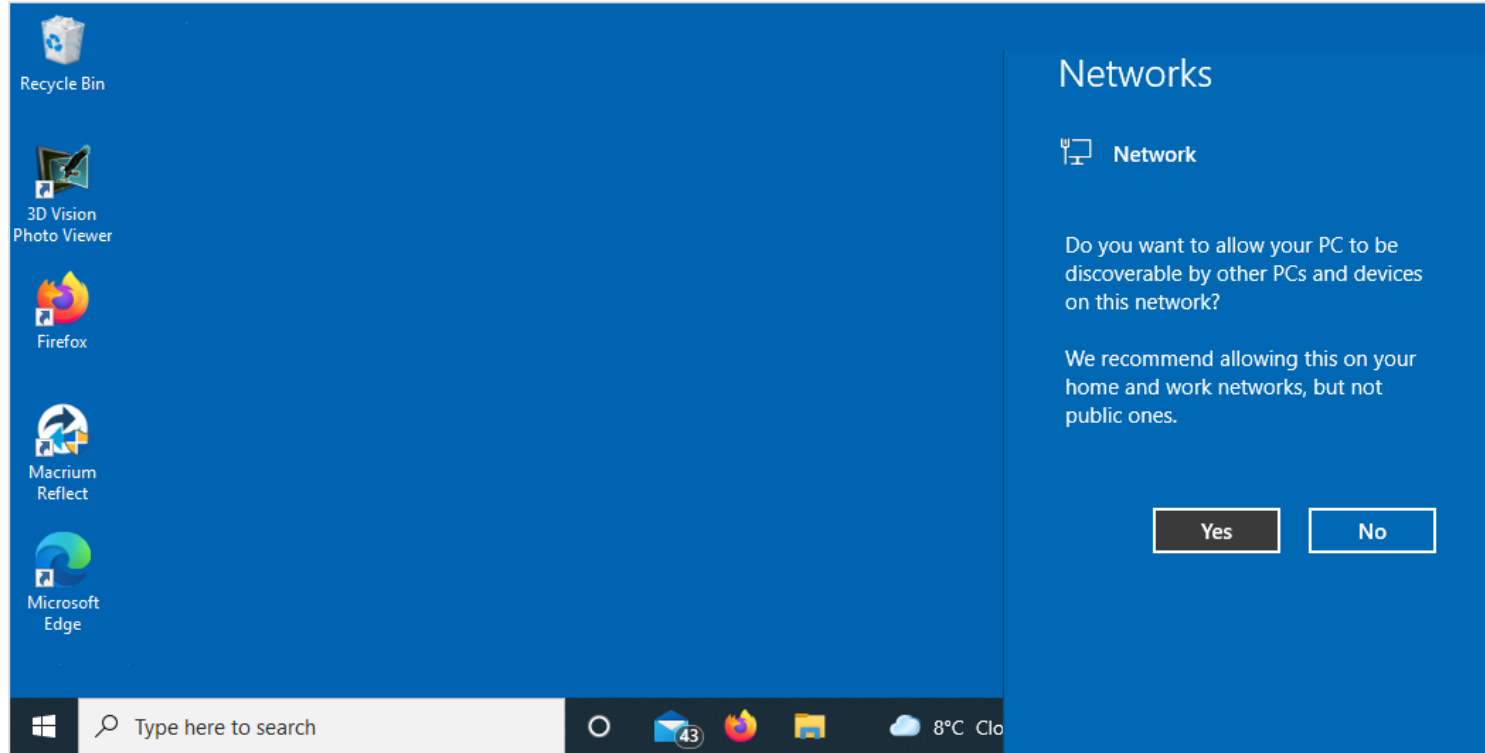


Screenshot courtesy of Microsoft.

Network Location (Slide 1 of 3)

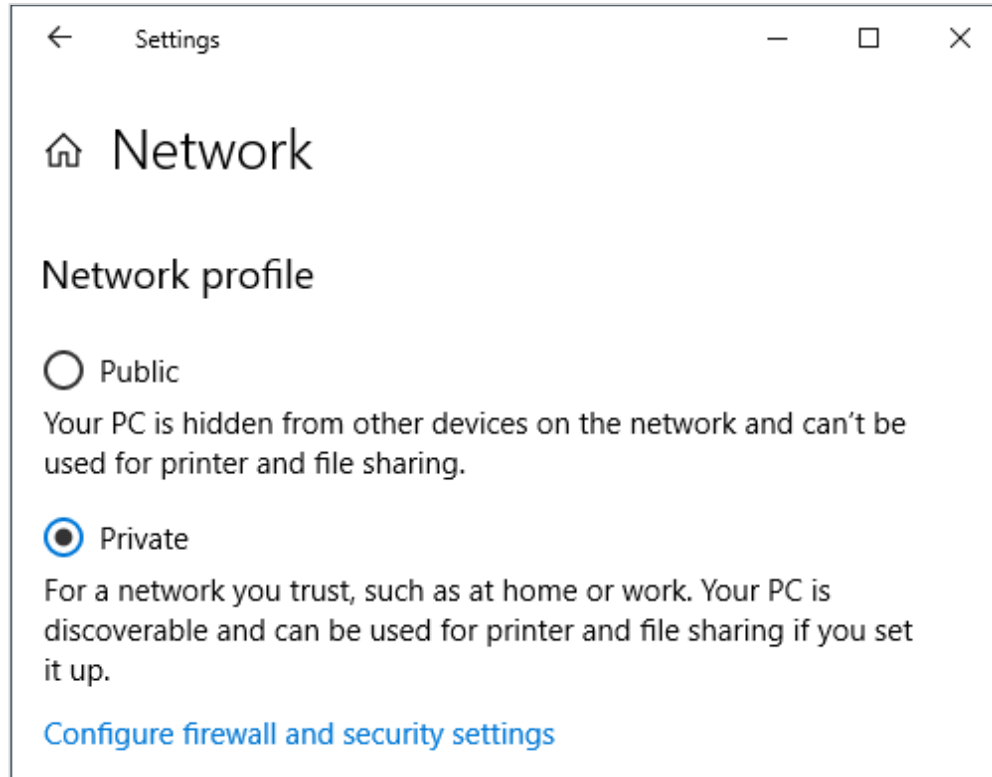
- Firewall controlled by Windows Defender Firewall
 - Unless third party firewall is installed
- Public, private, domain profiles
 - Public system
 - Not discoverable on network and sharing disabled
 - Private system
 - Discoverable on network and file and printer sharing
 - Domain
 - Used in enterprise environments
 - Firewall managed through a group policy

Network Location (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Network Location (Slide 3 of 3)



Screenshot courtesy of Microsoft.

Windows Defender Firewall Configuration (Slide 1 of 3)



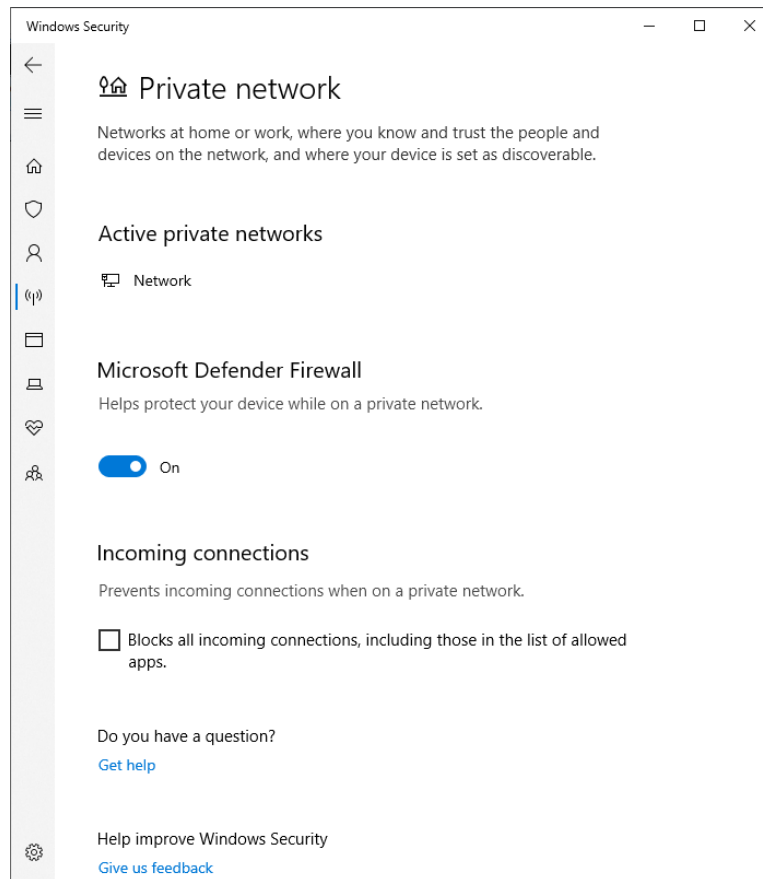
Windows Security Center

Set firewall and network protection



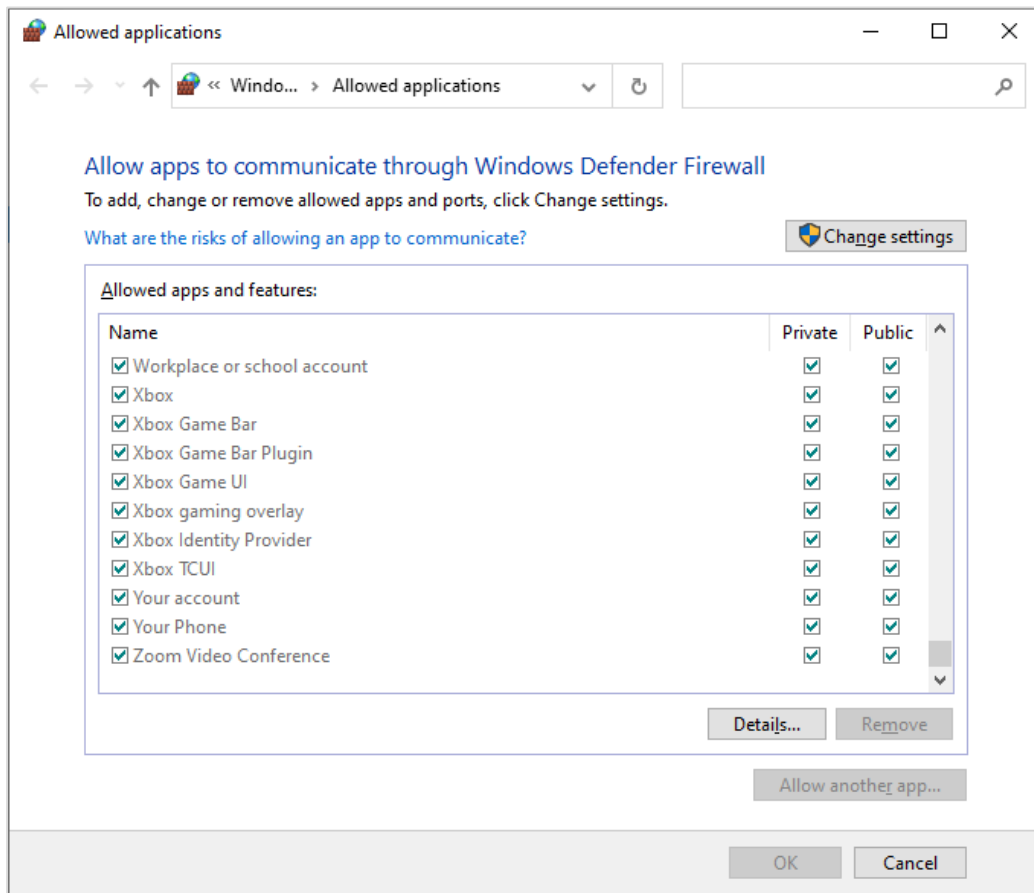
Allow or Block program or communication protocols

Windows Defender Firewall Configuration (Slide 2 of 3)



Screenshot courtesy of Microsoft.

Windows Defender Firewall Configuration (Slide 3 of 3)



Screenshot courtesy of Microsoft.

VPN and WWAN Connection Types (Slide 1 of 3)



Connects components together via a secure encrypted connection



Various options are available for VPN type



Third-party VPN apps for specific software

NordVPN, ExpressVPN, etc.



Wireless wide area networks

Cellular access card
Metered connection setting

VPN and WWAN Connection Types (Slide 2 of 3)

Settings

Add a VPN connection

VPN provider
Windows (built-in)

Connection name
COMPTIA

Server name or address
198.51.100.1

VPN type

- Automatic
- Point to Point Tunneling Protocol (PPTP)
- L2TP/IPsec with certificate
- L2TP/IPsec with pre-shared key
- Secure Socket Tunneling Protocol (SSTP)
- IKEv2

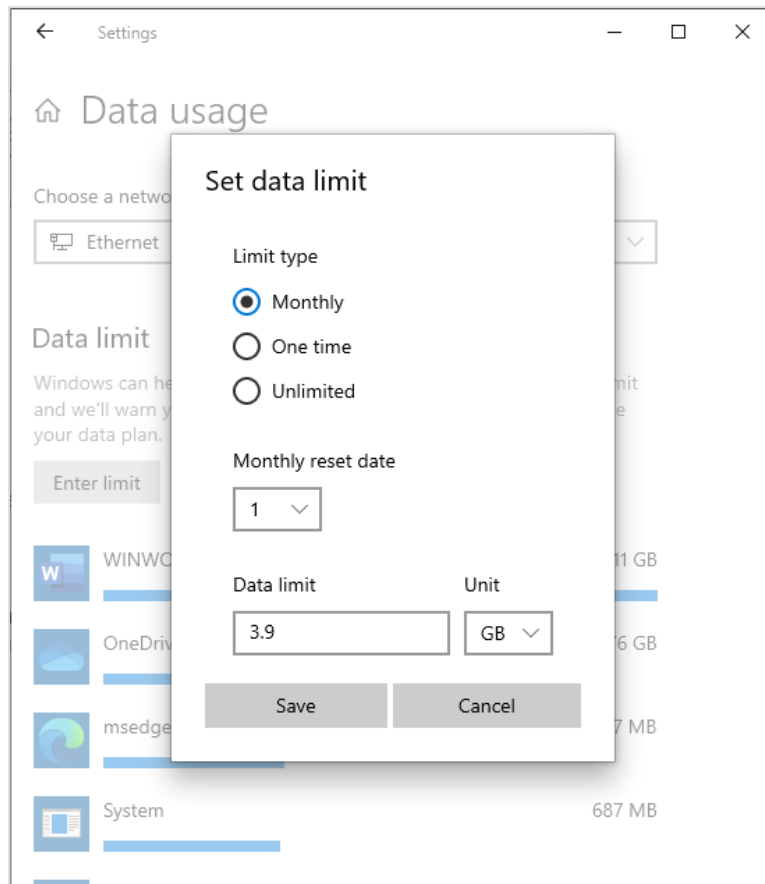
Password (optional)

☒ Remember my sign-in info

Save Cancel

Screenshot courtesy of Microsoft.

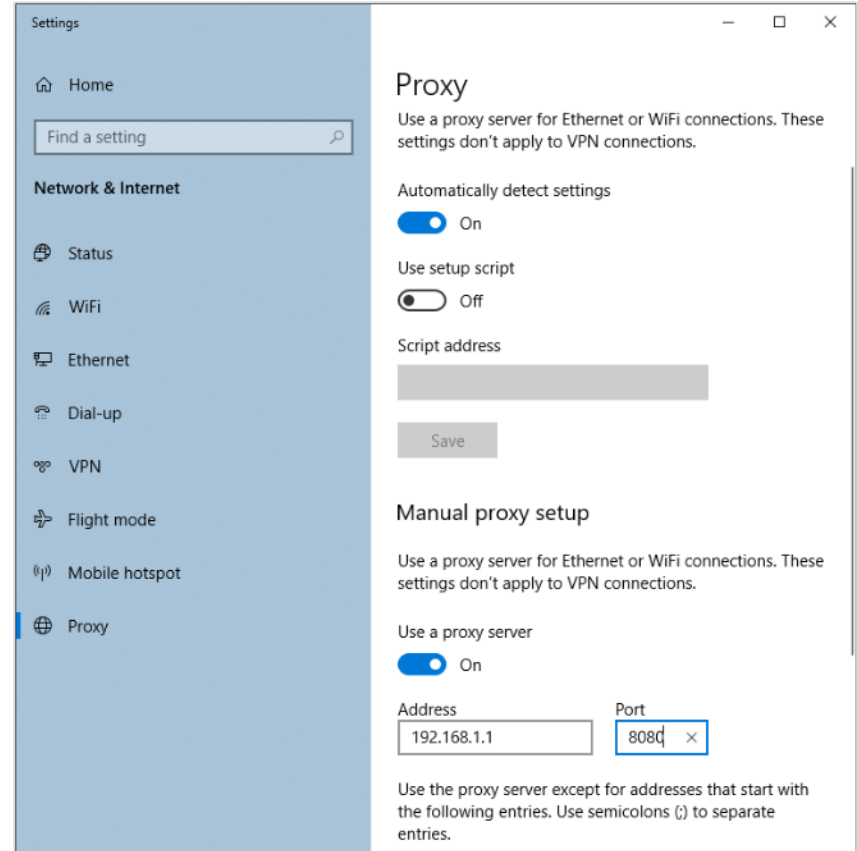
VPN and WWAN Connection Types (Slide 3 of 3)



Screenshot courtesy of Microsoft.

Proxy Settings

- Improves performance and security
- Saves bandwidth through local cache of web content
- IP and port of proxy server
 - Port 8080 is common



Lab Activity

- Lab: Configure a VPN Connection
(available in CertMaster Learn and CertMaster Perform)
- Lab: Local Firewall Settings
(available in CertMaster Learn and CertMaster Perform)
- Lab: Configure TCP/IP Settings on Windows 10
(available in CertMaster Learn and CertMaster Perform)
- Lab: Configure TCP/IP Settings on Windows 11
(available in CertMaster Learn and CertMaster Perform)
- Applied Live Lab: Manage Network and Security Settings
(available in CertMaster Perform and CertMaster Labs)

Lab Activity

- Challenge Live Lab: Manage Windows Workstations
(available in CertMaster Perform and CertMaster Labs)
- Challenge Live Lab: Set Up Windows Workstations
(available in CertMaster Perform and CertMaster Labs)

Lesson Review: Windows Networking

- Windows Network Connection Types
- IP Addressing Schemes
- Windows Client Configuration
- Network Location
- Windows Defender Firewall Configuration
- VPN and WWAN Connection Types
- Proxy Settings